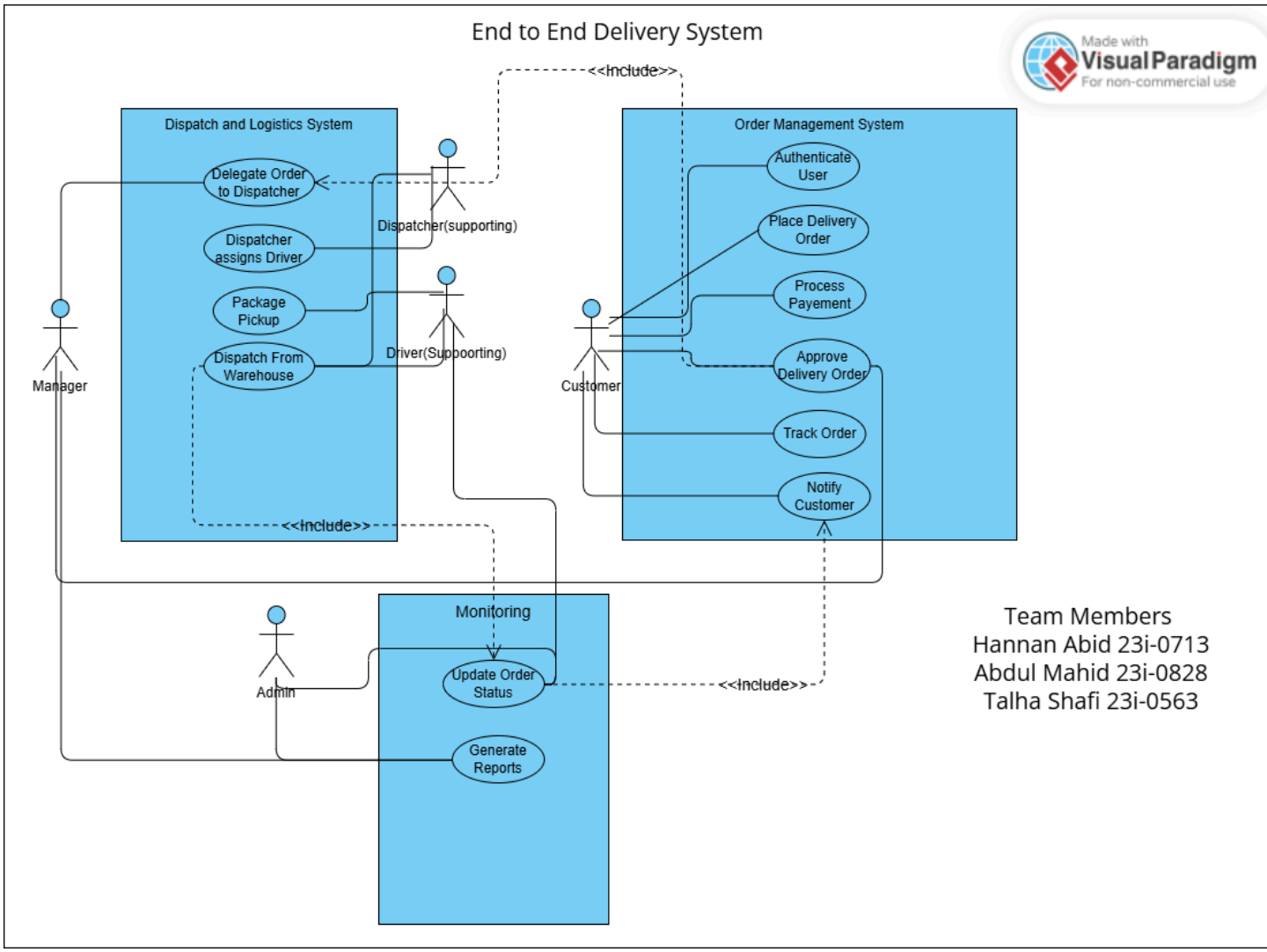
USE CASE DIAGRAM:

END TO END DELIVERY SYSTEM



Details of all USE-CASES

1. Authenticate User

Use Case Name: Authenticate User
Scope: Delivery Management System
Level: User goal
Primary Actor: User (Customer, Manager, Dispatcher, or Rider)
Stakeholders and their Interests:

- User: Wants secure and convenient access.
- System Administrator: Wants only authorized users to access the system.

Preconditions:

- User has registered an account with valid credentials.
- System is online and functional.

Postconditions:

- User is successfully logged in or authentication fails with appropriate error messages.

Main Success Scenario:

1. User navigates to the login page.
2. User enters username and password.
3. User clicks the “Login” button.
4. System verifies that the username exists.

5. System checks that the password matches.
6. System verifies the account status.
7. System grants access and displays the appropriate dashboard.

Extensions:

- If credentials are invalid then system displays error and user retries or requests password reset.
- If account is locked due to multiple failed attempts then user contacts admin.

2. Place Delivery Order and Attach Picture

Use Case Name: Place Delivery Order
Scope: Delivery Management System
Level: User goal
Primary Actor: Customer
Stakeholders and their Interests:

- Customer: Wants to successfully place orders.
- Business: Wants accurate order details for timely delivery.

Preconditions:

- User is authenticated.
- User has items to deliver.

Postconditions:

- Delivery order is created in the system.
- Picture attached is stored in database or cloud storage.

Main Success Scenario:

1. Customer navigates to the “Place Order” page.
2. Customer selects items and enters delivery details.
3. Customer uploads any required picture.
4. Customer reviews the order summary.
5. Customer submits the order.
6. System validates all fields and picture format.
7. Manager approves the order by reviewing the picture of the shipment
8. System stores the order and confirms placement.

Extensions:

- If picture upload fails then system prompts retry.
- If order details are incomplete then system prompts for missing information.

3. Process Payment

Use Case Name: Process Payment
Scope: Delivery Management System
Level: User goal
Primary Actor: Customer
Stakeholders and their Interests:

- Customer: Wants secure and successful payment.
- Business: Wants payment confirmation before processing order.

Preconditions:

- Order has been placed.
- Payment method is valid and active.

Postconditions:

- Payment is successfully processed or declined.
- Payment record is stored in the system.

Main Success Scenario:

1. Customer selects an order to pay.
2. Customer chooses a payment method.
3. Customer enters payment details.
4. System validates payment information.
5. System processes the transaction via the gateway.
6. System receives confirmation from the gateway.
7. System updates order to “Paid” and notifies the customer..

Extensions:

- If payment is declined then system notifies user and allows retry or selection of another method.
- If payment gateway is unavailable then system notifies user.

4. Approve Delivery Order

Use Case Name: Approve Delivery Order
Scope: Delivery Management System
Level: User goal
Primary Actor: Manager
Stakeholders and their Interests:

- Manager: Wants to ensure only valid orders proceed.
- Customer: Wants timely approval of order.

Preconditions:

- Order has been placed and payment processed.

Postconditions:

- Order status updated to “Approved.”

Main Success Scenario:

1. Manager navigates to “Pending Orders.”
2. Manager selects an order to review.
3. Manager verifies payment and order details and picture of shipment.
4. Manager checks compliance with policies.
5. Manager approves the order.
6. System updates order status to “Approved.”
7. System notifies customer and triggers dispatch.

Extensions:

- If order details are incomplete then manager requests corrections.
- If order is rejected then system updates status to “Rejected” and notifies customer.

5. Manager Delegates Order to Dispatcher

Use Case Name: Delegate Order to Dispatcher
Scope: Delivery Management System
Level: User goal
Primary Actor: Manager
Stakeholders and their Interests:

- Manager: Efficient task delegation.
- Dispatcher: Receives clear order assignment.

Preconditions:

- Order is approved.

Postconditions:

- Order is assigned to a dispatcher.

Main Success Scenario:

1. Manager views approved orders awaiting assignment.
2. Manager selects an order to delegate.
3. Manager checks dispatcher availability.
4. Manager assigns the order to a dispatcher.
5. System records the assignment.
6. System notifies the dispatcher of the new order.
7. Dispatcher confirms receipt in the system.

Extensions:

- If no dispatcher is available then system queues order.
- If dispatcher rejects assignment then manager reassigns.

6. Dispatcher Assigns Pickup Rider

Use Case Name: Assign Pickup Rider
Scope: Delivery Management System
Level: User goal
Primary Actor: Dispatcher
Stakeholders and their Interests:

- Dispatcher: Efficient delivery scheduling.

- Pickup Rider: Clear instructions for pickup.
- Customer: Timely collection.

Preconditions:

- Order is assigned to dispatcher.

Postconditions:

- Pickup rider assigned.
- Order status updated.

Main Success Scenario:

1. Dispatcher views assigned orders requiring pickup.
2. Dispatcher selects an order.
3. Dispatcher checks pickup rider availability.
4. Dispatcher assigns the order to a rider.
5. System updates the order with rider details.
6. System notifies the rider of assignment.
7. Rider confirms assignment in the system.

Extensions:

- If no riders are available then system schedules next available.
- If rider rejects then dispatcher reassigns.

7. Pickup Rider Collects Parcel from Customer

Use Case Name: Collect Parcel
Scope: Delivery Management System
Level: User goal
Primary Actor: Pickup Rider
Stakeholders and their Interests:

- Pickup Rider: Accurate and efficient collection.
- Customer: Parcel collected on time.
- Business: Parcel enters delivery process.

Preconditions:

- Pickup rider assigned.
- Customer is ready with parcel.

Postconditions:

- Parcel collected.
- Order status updated to “In Warehouse” or “Ready for Dispatch.”

Main Success Scenario:

1. Rider receives assignment notification.
2. Rider navigates to the customer’s location.
3. Rider verifies customer identity and parcel details.
4. Rider collects the parcel.
5. Rider confirms collection in the system.
6. System updates order status to “In Warehouse” or “Ready for Dispatch.”
7. System notifies the customer of successful collection.

Extensions:

- If customer unavailable then reschedule.
- If parcel is incomplete or damaged then rider reports issue.

8. Dispatch from Warehouse

Use Case Name: Dispatch from Warehouse
Scope: Delivery Management System
Level: User goal
Primary Actor: Warehouse Staff / Dispatcher

Stakeholders and their Interests:

- Warehouse Staff: Efficient inventory handling and parcel dispatch.
- Delivery Rider: Receives clear assignment for delivery.
- Business: Accurate stock management and smooth operations.

Preconditions:

- Parcel collected from pickup.
- Inventory is updated and parcel is ready at warehouse.

Postconditions:

- Parcel is ready for delivery.
- Delivery rider assigned.
- Order status updated to “Out for Delivery.”

Main Success Scenario:

1. Warehouse staff receives the parcel.
2. Staff verifies order details and updates warehouse inventory.
3. Staff prepares the parcel for delivery.

4. Staff assigns the parcel to a delivery rider based on availability.
5. System updates the order status to “Out for Delivery.”
6. System notifies the delivery rider with delivery instructions.
7. Delivery rider confirms assignment in the system.

Extensions:

- If inventory discrepancy occurs then staff reports the issue.
- If parcel is damaged then staff informs manager.
- If rider is unavailable then system schedules the next available rider.

9. Notify Customer

Use Case Name: Notify Customer
Scope: Delivery Management System
Level: User goal
Primary Actor: System / Warehouse Staff / Dispatcher

Stakeholders and their Interests:

- Customer: Wants timely and accurate updates about their order.
- Business: Enhances transparency and trust with customers.

Preconditions:

- Parcel is dispatched and order status is updated to “Out for Delivery.”

Postconditions:

- Customer receives notification of parcel status and expected delivery.

Main Success Scenario:

1. System detects that a parcel has been dispatched.
2. System retrieves customer contact information.
3. System generates a notification message (SMS, email, or app notification).
4. System sends notification to the customer.
5. System logs the notification for record and auditing.
6. Customer receives notification and can track order.
7. System updates notification status to confirm successful delivery of message.

Extensions:

- If notification fails then system retries sending automatically.
- If contact information is missing or invalid then staff manually contacts the customer.

10. Track Order

Use Case Name: Track Order
Scope: Delivery Management System
Level: User goal
Primary Actor: Customer
Stakeholders and their Interests:

- Customer: Real-time order status.
- Business: Transparency builds trust.

Preconditions:

- Order exists and has valid status.

Postconditions:

- Customer views current order location/status.

Main Success Scenario:

1. Customer navigates to “Track Order.”
2. Customer selects the order to track.
3. System retrieves the current status and location.
4. System displays the delivery route and estimated arrival.
5. System shows history of previous status updates.
6. System updates tracking information in real-time.
7. Customer can contact support if there are issues.

Extensions:

- If order ID is invalid then system displays error.
- If system cannot retrieve location then display last known status.

11. Update Order Status

Use Case Name: Update Order Status
Scope: Delivery Management System
Level: User goal
Primary Actor: Rider / Dispatcher / Warehouse Staff
Stakeholders and their Interests:

- Customer: Accurate status updates.
- Business: Operational transparency.

Preconditions:

- Order is in process.

Postconditions:

- Order status updated in system.

Main Success Scenario:

1. Actor logs into the system.
2. Actor navigates to assigned orders.
3. Actor selects an order to update.
4. Actor enters the current status of the order.
5. System validates and records the status.
6. System notifies the customer of the update.
7. System updates dashboards and reports with the new status.

Extensions:

- If status update is invalid then system prompts correction.

12. Generate Report

Use Case Name: Generate Report
Scope: Delivery Management System
Level: User goal
Primary Actor: Manager / Admin
Stakeholders and their Interests:

- Manager/Admin: Insight into performance, deliveries, payments.
- Business: Data-driven decisions.

Preconditions:

- System has relevant data.
- User has proper privileges.

Postconditions:

- Report generated and stored/exported.

Main Success Scenario:

1. Manager or admin logs into the reporting section.
2. User selects the report type.
3. System retrieves relevant data from the database.
4. System aggregates and formats the data.
5. System generates and displays the report.
6. User reviews, downloads, or exports the report.
7. System logs report generation for auditing purposes.

Extensions:

- If no data available then system displays “No records found.”
- If export fails then system prompts retry.